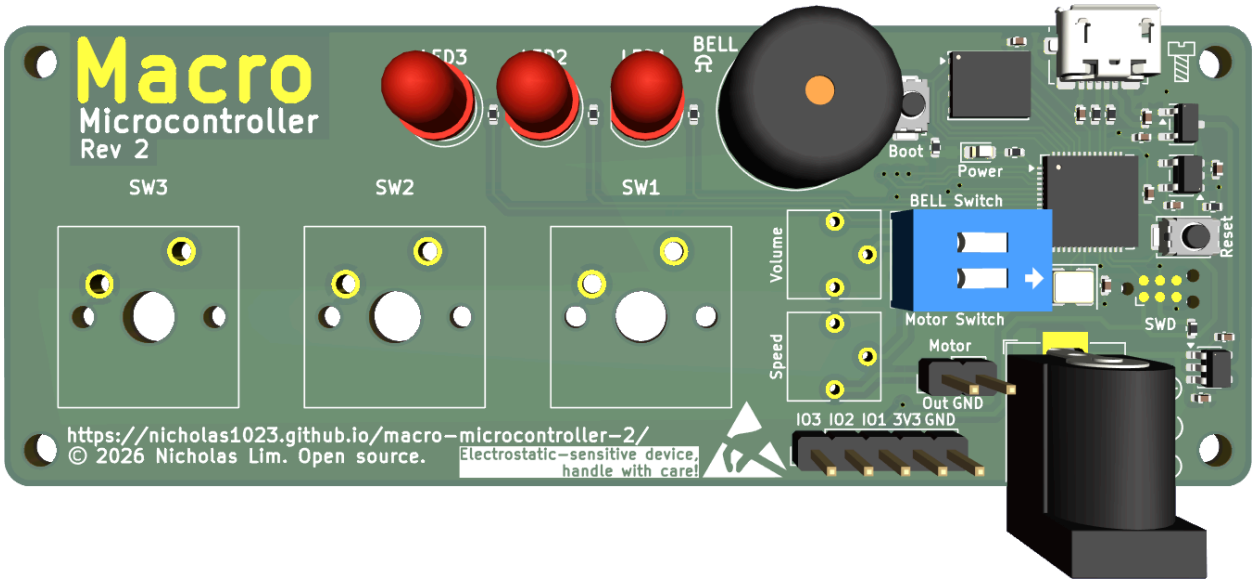
Macro Microcontroller 2 Datasheet



Overview

Macro Microcontroller 2 is an RP2040 microcontroller development board with an integrated macropad that lets you control GPIOs with keyboard keys on a single PCB, plus a custom BASIC interpreter and disk operating system.

Specifications

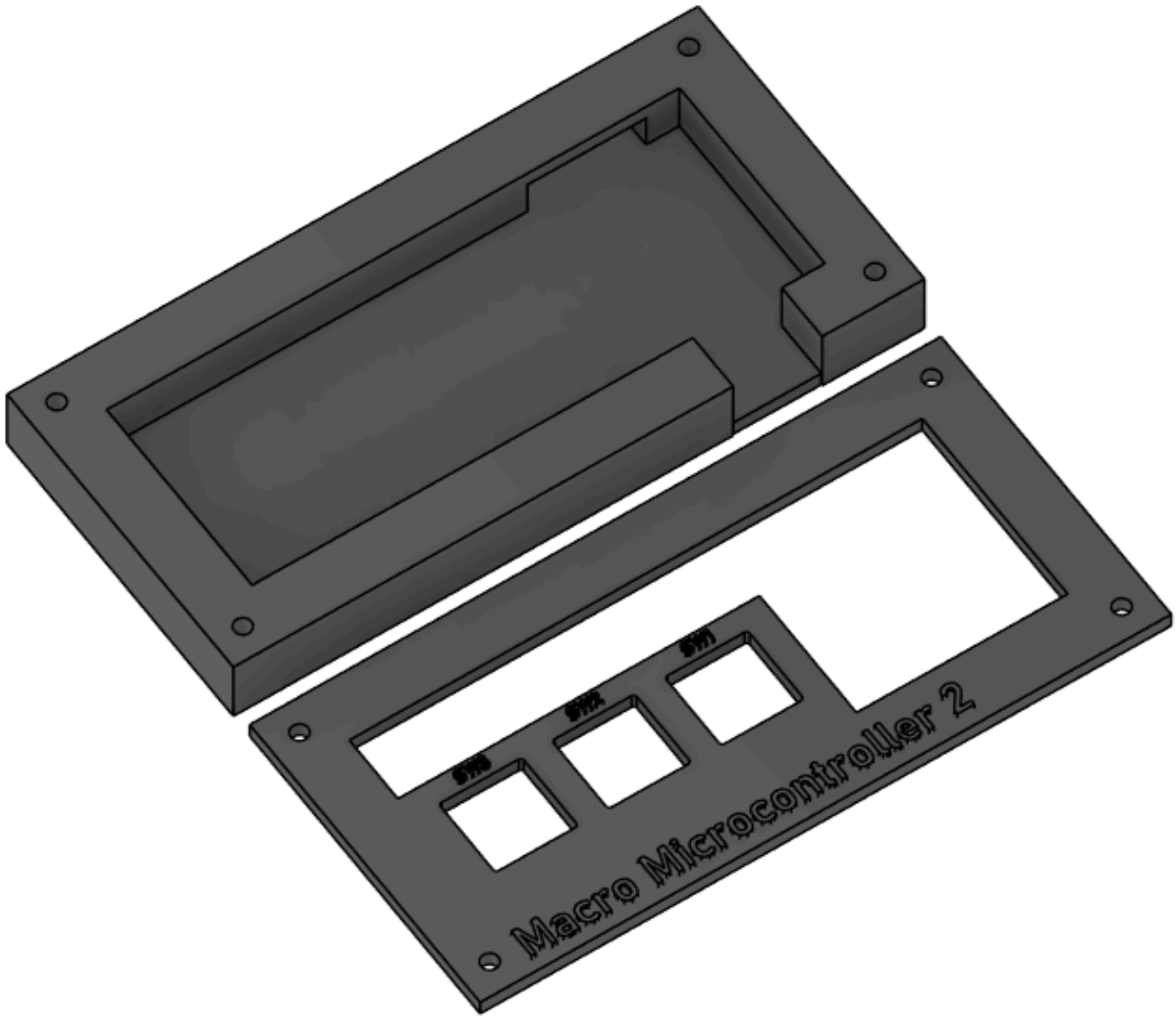
Microcontroller Unit: RP2040
Power: 5V input via data port or barrel jack, 3.3V input via GPIO pin headers, maximum logic voltage is 3.3V.
Dimensions: 96.52 × 35.56 mm
Flash Memory: 16 MB
Mounting Hole Dimensions: 2.5mm diameter

Precautions

- Do not apply logic voltages exceeding 3.3V to any GPIO or inputs. This may cause onboard components to be damaged.
- Use only one power source at a time, either via the data port, barrel jack or 3V3 and GND pins to power the board.

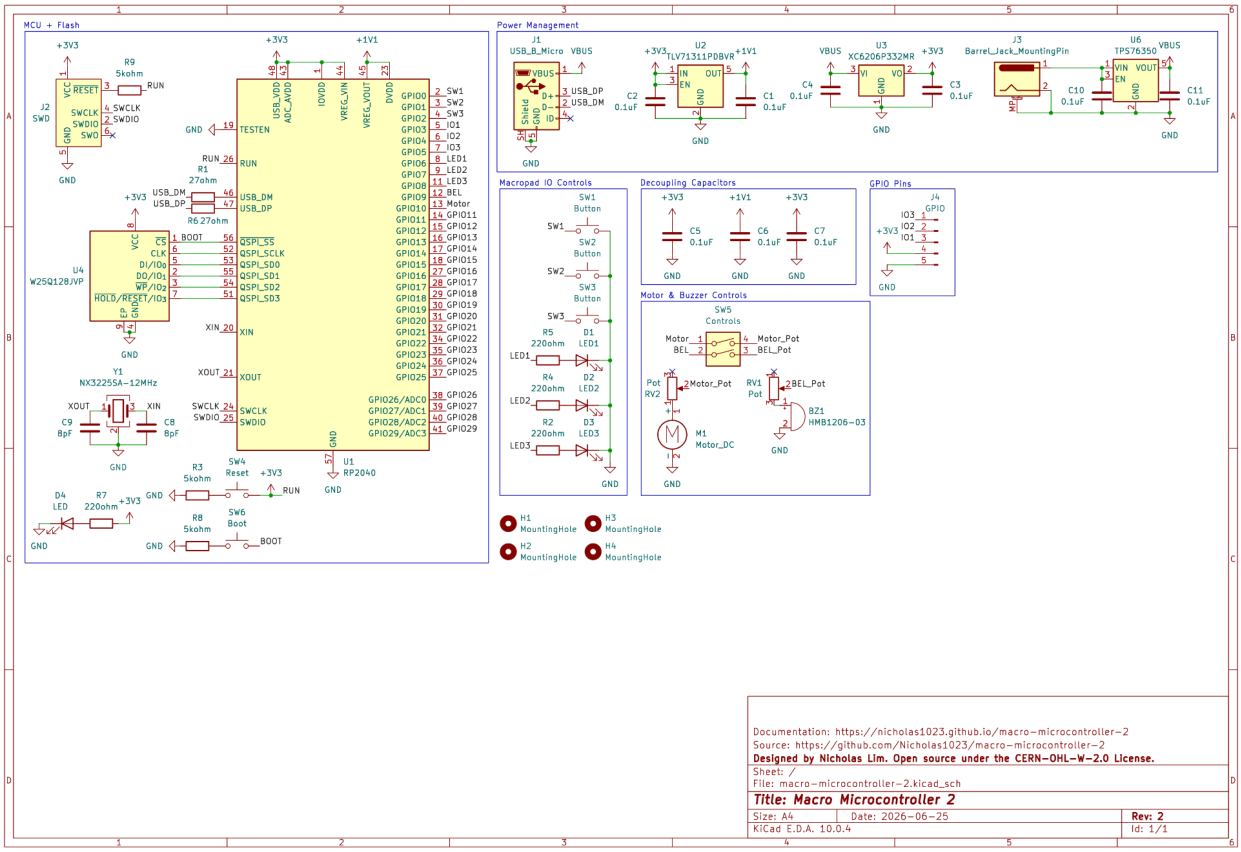
- Only the data port and barrel jack accepts a maximum voltage of 5V. Onboard Low Dropout (LDO) Regulators provide regulated 3.3V when powered by either inputs only. The 3V3 pin provides a separate power connection and bypasses these input regulators.
- This is an electrostatic-sensitive device. Handle the board only by the sides and avoid touching components and pins.
- Do not handle with wet hands.

Enclosure

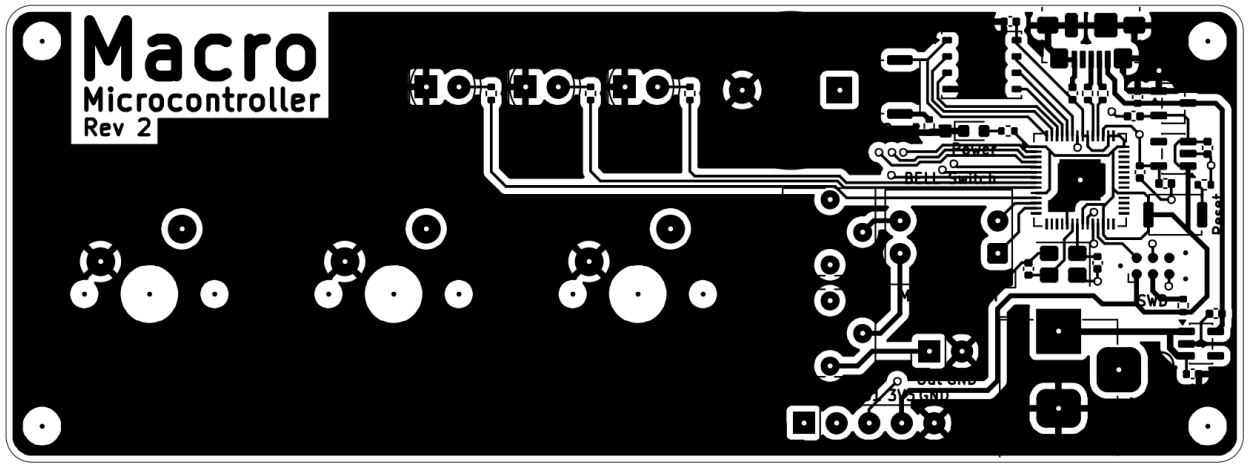


This enclosure is 3D-printable and is available for download from https://github.com/Nicholas1023/macro-microcontroller-2/releases/download/v0.0.1/macro-microcontroller-2-enclosure.stl

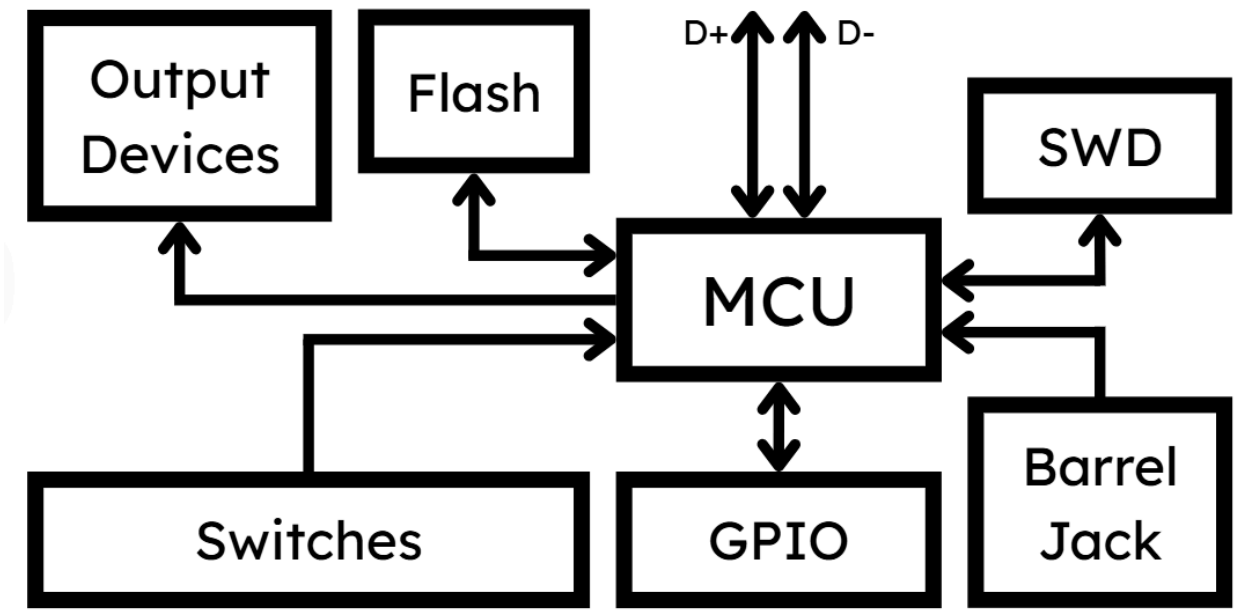
Schematics



PCB Routing and Layout



Block Diagram



Pinout

Macro Microcontroller 2 provides a few IO pinouts for connecting peripherals.

- DC Motor (MOTOR)
- 2.54mm GPIO pin headers (IO3, IO2, IO1, 3.3V, GND)

Macro Microcontroller 2 Pin #	Corresponding RP2040 Pin #
IO1	GPIO3
IO2	GPIO4
IO3	GPIO5

Built-in IO Devices

- Key switches (SW1, SW2, SW3)
- LEDs (LED1, LED2, LED3)
- Buzzer (BELL)

Bill of Materials

Qty	Reference Designator	Manufacturer Part #
1	BZ1	HMB1206-03
1	J3	DC-005 2.0
1	SW5	DS-02RP
1	M1	2.54 1x2 Pin Header
1	J4	2.54 1x5 Pin Header
1	J1	473460001
9	C1, C10, C11, C2, C3, C4, C5, C6, C7	0.1µF 0402
1	U1	RP2040
1	U3	XC6206P332MR
1	U2	TLV71311PDBVR
4	R2, R4, R5, R7	220Ω 0402
1	D4	NCD0603B1
2	C8, C9	8pF 0402
1	U4	W25Q128JVPIQ
3	D1, D2, D3	333-2SURD/S530-A3
1	U6	TPS76350DBVR
1	Y1	NX3225SA-12MHz-STD-CSR-6
2	R1, R6	27Ω 0402
3	R3, R8, R9	5kΩ 0402
2	RV1, RV2	T73YP202KT20
3	SW1, SW2, SW3	MX1A-E1NA
2	SW4, SW6	SKRKAEE020

Macro Microcontroller DOS

Macro Microcontroller Disk Operating System (DOS) is a simple command-line operating system providing basic programs for Macro Microcontroller 2.

Installation

1. Download macro-microcontroller-dos-0.0.3.uf2. This release includes Macro Microcontroller BIOS, A basic BIOS-like firmware, which performs POST routines, reports information on the device and boot the DOS.

   SHA-256 Checksum:
   `e0ba748dbac0241309680e81cc556e139e9c126062bc54f37d4ad093bcaae639`
2. Plug in Macro Microcontroller 2 while holding the Boot button. A storage drive will appear on your computer.
3. Move macro-microcontroller-dos-0.0.3.uf2 into the storage drive. Your development board is now running Macro Microcontroller BIOS and DOS.
4. Install a serial terminal application on your computer and connect it to the board. If successful, you should see the following:

```
 _ _
| \/ | Macro Microcontroller BIOS Version 0.0.2.
|_\/_| Copyright (C) 2025-2026 Nicholas Lim.

Device: Macro Microcontroller 2 (RP2040 @ 125MHz)
Serial Number: ****************
RAM Free: 254K OK
Serial Status: Connected

Press SW1 to enter Macro Microcontroller BASIC.
Press SW2 to enter BOOTSEL mode.

Booting system...

Macro Microcontroller Disk Operating System Version 0.0.3.
Copyright (C) 2026 Nicholas Lim.
Drive A>
```

Commands and Programs

- `HELP`: Launches the help guide.
- `CALCULATE`: Performs basic operations on 2 numbers. Usage: `CALCULATE [Number] [+ - * / ^ root] [Number]`
- `ECHO`: Echoes text.

- `PROGLIST`: Show program listing.
- `SYSTEM`: Starts another Macro Microcontroller DOS console.
- `EXIT`: Stops all processes for power off.
- `OUT`: Turns on a specified GPIO for 1 second. Usage: `OUT [LED# IO# BELL MOTOR]`
- `VERSION`: Shows the release version.
- `BOOT`: Reboots the system. Usage: `BOOT -m [BASIC BOOTSEL]*`
- `DEVINFO`: Displays device and memory information.
- `UPTIME`: Shows device uptime.

Predefined Values for Parameters

- `IO*`: Set GPIO pins 1, 2 or 3 to high.
- `LED*`: Set LEDs 1, 2 or 3 to high.
- `BELL`: Sounds the builtin buzzer.
- `MOTOR`: Sets motor pins to high.
- `BASIC`: Reboot in BASIC interpreter mode.
- `BOOTSEL`: Reboot in BOOTSEL mode.

Macro Microcontroller BASIC

Macro Microcontroller BASIC is a custom BASIC interpreter written in C with low-level hardware communication, providing a programming environment via a serial terminal.

Installation

1. Download macro-microcontroller-basic-0.0.3.uf2.

   SHA-256 Checksum:
   `96d4aa58d61d796067c1668cc37896fa968a55dceea188e50f26e067d116e99f`
2. Plug in Macro Microcontroller 2 while holding the Boot button. A storage drive will appear on your computer.
3. Move macro-microcontroller-basic-0.0.3.uf2 into the storage drive. Your development board is now running Macro Microcontroller BASIC.
4. Install a serial terminal application on your computer and connect it to the board. If successful, you should see the following:

   ```
   Macro Microcontroller BASIC Version 0.0.3.
   Copyright (C) 2025-2026 Nicholas Lim.
   ****** bytes of RAM free.
   Ready.
   ```

```
10
```

5. If not, try sending `PRINT "It works!"`. If "It works!" is sent back to the serial terminal, you have successfully installed Macro Microcontroller BASIC.

Statements

- `PRINT`: Outputs text to the serial terminal.
- `LET`: Assigns a value (integer, float or string) to a variable.
- `INPUT`: Assigns a value (integer, float or string) to a variable via user input.
- `GPIO`: Turns on a specified GPIO pin for 1 second.
- `LSVAR`: Outputs variable name and their value to the serial terminal.
- `END`: Clears variable and reset line numbers.
- `EXIT`: Puts the device into sleep/power off mode.
- `KEY`: Pass GPIO controls to the macropad.
- `REM`: Add comments to your program.

Predefined Values

- `IO*`: Set GPIO pins 1, 2 or 3 to high.
- `LED*`: Set LEDs 1, 2 or 3 to high.
- `BELL`: Sounds the builtin buzzer.
- `MOTOR`: Sets motor pins to high.
- `RND`: Generates a random floating-point integer less than 1.

For more information on Macro Microcontroller 2, visit:
https://github.com/Nicholas1023/macro-microcontroller-2 (Repository)
https://nicholas1023.github.io/macro-microcontroller-2 (Documentation site)

Licensing

Hardware files ("PCB Files", "Printing Files", "Production Files", "schematics.pdf") are licensed under the terms of the CERN-OHL-W-2.0 License from 9 September 2026 onwards. Versions prior to this date remain available under the terms of the CERN-OHL-P-2.0 License. See LICENSE.txt.

Documentation files ("docs", "Assets", "JOURNAL.md" and this datasheet) are licensed under the terms of the MIT License. See docs/LICENSE.txt.

macro-microcontroller-basic.c, macro-microcontroller-bios.c, macro-microcontroller-bios+dos.c are licensed under the terms of the MIT License.

firmware-template.c is licensed under the terms of the MIT-0 License. example.bas is marked CC0 1.0.

For further details on legal information and attribution, please refer to the Legal section in README.md.